

Direction	Ether Type	IP Protocol	Port Range	Remote IP Prefix	Remote Security Group	Description	Actions
Egress	IPv4	Any	Any	0.0.0.0/0	-	-	Delete Rule
Egress	IPv6	Any	Any	:::0	-	-	Delete Rule
Ingress	IPv4	ICMP	Any	0.0.0.0/0	-	-	Delete Rule
Ingress	IPv4	TCP	22 (SSH)	0.0.0.0/0	-	-	Delete Rule
Ingress	IPv4	TCP	80 (HTTP)	0.0.0.0/0	-	-	Delete Rule
Ingress	IPv4	TCP	179	0.0.0.0/0	-	Calico BGP network	Delete Rule
Ingress	IPv4	TCP	2379 - 2380	0.0.0.0/0	-	etcd server client API	Delete Rule
Ingress	IPv4	TCP	6443	0.0.0.0/0	-	Kubernetes API Server	Delete Rule
Ingress	IPv4	TCP	6781 - 6784	0.0.0.0/0	-	Weave Net	Delete Rule
Ingress	IPv4	TCP	8080	0.0.0.0/0	-	Jenkins Port	Delete Rule
Ingress	IPv4	TCP	9099	0.0.0.0/0	-	Calico felix (health check)	Delete Rule
Ingress	IPv4	TCP	10250	0.0.0.0/0	-	Kubelet API	Delete Rule

Figure 3: Kubernetes cluster security group for all the ingress and egress

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
DevOps-Tools	ubuntu20.04-x86_64-disk	10.0.5.135, 172.24.4.92	m1.medium	nfv-k8s	Active	nova	None	Running	0 minutes	Create Snapshot
K8s-Worker	ubuntu20.04-x86_64-disk	10.0.5.15, 172.24.4.213	m1.medium	nfv-k8s	Active	nova	None	Running	3 minutes	Create Snapshot
K8s-Master	ubuntu20.04-x86_64-disk	10.0.5.91, 172.24.4.182	m1.large	nfv-k8s	Active	nova	None	Running	4 minutes	Create Snapshot

Figure 4: Various VM instances created over OpenStack for the Kubernetes cluster

public IP address to them for it to enable connection to the internet. The flavour in the figure showcases the configurations and the computation resources provided to these instances.

The OpenStack dashboard shows comprehensive information about the various resources being utilised within the working project. In Figure 5, we see all the instances running and the number of virtual CPUs and RAM allocated to them. Apart from this, other

network details are visible too, where the number of floating IPs attached to the instances is visible along with the number of security groups and their rules created within the working project. This information is helpful for the administrator to have a look over the metadata related to all the services provided by OpenStack.

Ansible completed all its allocated tasks by deploying the Kubernetes objects over the cluster sitting on

top of OpenStack. Figure 6 shows all the configured objects running on the Kubernetes cluster after the successful deployment. The Kubernetes configuration files also deployed certain services within the cluster so that the NFV application could be invoked from outside the network.

Here, in this cluster for the experimentation, the configuration files call for one active pod of